

PRODUCT PROPOSAL

CertChain

Blockchain-Powered Document & Certificate Verification System

Proof of Concept — Version 1.0

Confidential & Proprietary

May 2026

1. Executive Summary

CertChain is a blockchain-powered platform for issuing, managing, and instantly verifying tamper-proof digital certificates, degrees, licences, contracts, and credentials. Every document issued through CertChain is anchored to an Ethereum-compatible blockchain as a cryptographic hash — making it permanently verifiable by anyone, anywhere, without relying on a central authority or calling the issuing institution.

The platform addresses a documented, multi-sector crisis: credential fraud in education, slow and error-prone licence verification in HR and legal, and costly manual authentication in financial and government services. CertChain replaces paper certificates, email PDFs, and siloed institutional registries with a single, open verification standard.

This document presents the full business case, technical architecture, feature set, commercial model, and implementation roadmap for CertChain v1.0 — designed both as a college-level Proof of Concept and a market-ready product for enterprise clients.

2. Problem Statement

Document and credential fraud is a global epidemic with direct commercial consequences:

- Degree & certificate fraud — a 2024 NASSCOM study estimated that 20–30% of engineering resumes in India carry inflated or fabricated academic credentials. Companies spend weeks and thousands of rupees per candidate on background verification.
- Slow licence verification — law firms, hospitals, and financial institutions manually call regulators or institutions to verify professional licences and bar enrolments, taking 3–10 business days per check and often failing entirely for foreign credentials.
- Contract authenticity disputes — paper and PDF contracts are easily tampered with after signing, leading to costly litigation over document authenticity.
- Government certificate fraud — fake birth certificates, caste certificates, and income certificates are a persistent issue in government welfare and admission processes.
- No universal standard — every institution uses a different format (PDF, physical paper, proprietary portals), making cross-border or cross-institution verification nearly impossible.

Existing partial solutions — LinkedIn badges, DigiLocker, institutional portals — are either non-cryptographic (easily faked), siloed (not interoperable), or inaccessible to third-party verifiers without institutional permission. The market needs an open, cryptographically verifiable, institution-agnostic standard.

3. Solution Overview

CertChain works in two simple steps: the issuer registers a document hash on-chain; the verifier checks the hash. No intermediary, no phone call, no waiting.

Core principle: CertChain never stores the document itself on-chain. It stores only a SHA-256 hash of the document. The issuer retains the original. Any third party can verify authenticity in under three seconds.

3.1 Issuance Flow

- Authorised issuer (university, employer, government body) uploads the document or fills a structured template in the CertChain dashboard.
- The platform computes a SHA-256 hash of the document content + metadata (recipient name, issue date, expiry if applicable, issuer ID).
- A CertificateRegistry smart contract stores the hash on Polygon PoS mainnet with the issuer's wallet signature and a timestamp.
- CertChain generates a unique Certificate ID (UUID mapped to on-chain token) and a shareable verification link / QR code.
- The recipient receives a tamper-proof PDF or digital badge with the Certificate ID and QR embedded.

3.2 Verification Flow

- Verifier (recruiter, regulator, court, bank) receives the document and Certificate ID — or simply scans the QR code.
- CertChain's public verification portal computes the hash of the presented document and queries the smart contract.
- Result returned in under 3 seconds: VALID (hash matches, issuer confirmed) or INVALID (tampered / not found).
- No account, no login, no fee required for verification. Fully public.

4. Technical Architecture

4.1 Smart Contract Layer (Solidity on Polygon)

- CertificateRegistry.sol — core contract. Functions: issueCertificate(bytes32 hash, string metadata), revokeCertificate(bytes32 certId), verifyCertificate(bytes32 certId) returns (bool, address issuer, uint256 timestamp).
- IssuerRegistry.sol — whitelist of authorised issuers. Only admin-approved wallets can call issueCertificate. Prevents impersonation.

- CertNFT.sol — optional ERC-721 module. Mints a non-transferable (soulbound) NFT to the recipient's wallet as a verifiable on-chain credential. Supports OpenSea metadata standard for interoperability.
- Deployed on Polygon PoS — EVM-compatible, ~\$0.001 per transaction, 2-second finality.

4.2 Backend & API Layer

- Node.js / Express REST API — handles document ingestion, hash computation, relayer transactions (gasless UX for issuers), and webhook delivery.
- IPFS / Pinata — optional encrypted document storage. Hash stored on-chain; full document pinned to IPFS for issuers who want permanent document backup.
- PostgreSQL — off-chain metadata index (certificate ID, recipient name, institution, course, expiry) for fast dashboard queries and bulk export.
- Gasless relay (ERC-2771 meta-transactions) — issuers never need ETH or MATIC; the platform absorbs gas costs and bundles them into the subscription fee.

4.3 Frontend

- Issuer Dashboard (React.js) — bulk certificate issuance via CSV upload, template builder, revocation management, analytics (issued / revoked / verified counts).
- Public Verification Portal — zero-login page. Paste Certificate ID or scan QR. Instant VALID / INVALID result with issuer name, issue date, and expiry.
- Recipient Wallet View — React Native mobile app. Recipients collect their certificates as soulbound NFTs. Share via link, QR, or LinkedIn Open Badges.
- MetaMask + WalletConnect — issuer wallet authentication. Verifiers require no wallet.

4.4 Integration & Standards

- Open Badges 3.0 (IMS Global) — certificates exportable as W3C Verifiable Credentials, compatible with LinkedIn, Credly, and HRMS platforms.
- DigiLocker API — certificates issued via CertChain can optionally be pushed to a recipient's DigiLocker for government-recognised digital storage.
- Webhook API — HRMS, ATS, and LMS platforms (Workday, Darwinbox, Moodle) can trigger certificate issuance directly via REST API.
- Bulk CSV issuance — universities can upload a graduation list and issue 10,000 certificates in a single transaction batch.

5. Use Cases & Target Verticals

Vertical	Document Types	Value Delivered
----------	----------------	-----------------

Education	Degrees, diplomas, marksheets, MOOCs	Eliminate fake certificates; instant BGV for recruiters; alumni self-service
Corporate HR	Experience letters, offer letters, appraisal records	Reduce BGV turnaround from 10 days to 3 seconds; prevent offer fraud
Legal & Compliance	Bar enrolments, court orders, notarised deeds	Tamper-proof legal records; instant authenticity check in dispute resolution
Healthcare	Medical degrees, NMC registration, CME credits	Hospitals verify doctor credentials in minutes; prevent practitioner fraud
Government	Birth/income/caste certificates, licences	Eliminate forgeries in welfare schemes; interoperability across states
Fintech / KYC	PAN, Aadhaar linkage proofs, KYC documents	Reusable KYC credentials; reduce onboarding cost by 60-80%

6. Feature Set

6.1 MVP — Phase 1 (POC Deliverable)

Feature	Description	Priority
Certificate Issuance	Hash document, sign with issuer wallet, store on-chain	P0
Public Verification Portal	Anyone can verify by Certificate ID or QR — no login	P0
QR Code Generation	Embed QR in issued certificate PDF for instant scan-to-verify	P0
Certificate Revocation	Issuer can revoke a certificate; verification shows REVOKED status	P0
Issuer Dashboard	Manage issued certs, view verification logs, revoke, download	P0
Bulk CSV Issuance	Upload CSV of recipients; issue hundreds of certs in one click	P1
Expiry / Validity Period	Set optional expiry date; verification flags expired credentials	P1

6.2 Scale-up — Phase 2 (Commercial)

- Soulbound NFT credentials — non-transferable ERC-721 token in recipient's wallet; verifiable on any EVM explorer.

- W3C Verifiable Credentials export — Open Badges 3.0 JSON-LD for LinkedIn and HRMS integration.
- DigiLocker push — one-click delivery to recipient's government DigiLocker account.
- White-label portal — client-branded verification URL (e.g., verify.iitbombay.ac.in).
- HRMS / LMS webhooks — Workday, Darwinbox, Moodle, Canvas trigger issuance via API.
- Analytics & audit trail — full log of who verified what certificate and when (GDPR-compliant hashed IPs).
- Multi-issuer consortium — e.g., all state universities share one registry; bar councils share one licence registry.

7. Commercial & Pricing Model

CertChain is offered as a B2B SaaS platform. Pricing is based on the volume of certificates issued per month (not verifications — verification is always free and public). On-chain gas fees on Polygon are absorbed into the platform subscription.

Tier	Starter	Growth	Enterprise
Monthly Price	₹10,000 / mo	₹35,000 / mo	Custom
Certs Issued / mo	Up to 500	Up to 10,000	Unlimited
Verifications	Unlimited (free)	Unlimited (free)	Unlimited (free)
Soulbound NFTs	No	Optional (+₹8k)	Included
White Label	No	Optional (+₹15k)	Included
API / Webhooks	No	Yes	Yes + SLA
DigiLocker Push	No	Yes	Yes
Support	Email	8x5 dedicated	24x7 + CSM

Additional revenue streams: one-time onboarding / integration fee (Rs.50k-2L for HRMS/LMS hookups); white-label licensing for system integrators reselling to universities; government tender revenue for state-level deployment; and a B2C verifier API for background verification companies (Rs.5 per API call).

8. Target Market & Client Profiles

8.1 Immediate Targets (0-12 months)

- Private engineering and management colleges (100-500 annually graduating students) — easiest POC conversion; compliance with NEP 2020 Academic Bank of Credits (ABC) mandate.
- Mid-size IT / ITES companies — 500-5,000 employee firms that issue experience letters manually and spend Rs.500-2,000 per candidate on BGV.
- Professional certification bodies — ICAI, ICSI, bar councils, nursing councils issuing 10,000+ certificates per year with zero digital verification infrastructure.

8.2 Scale Targets (12-36 months)

- State government departments — NIC integration for digitising state-issued certificates (income, caste, domicile) under the PM Gati Shakti / Digital India umbrella.
- NBFCs and banks — KYC document verification at onboarding; reusable credential layer reducing per-customer onboarding cost.
- Hospital networks — doctor credentialing across multi-city hospital chains; NMC registration verification.

9. Implementation Roadmap

Phase	Timeline	Deliverables
Phase 0 — POC	Weeks 1-3	Smart contracts on Sepolia testnet; React issuer dashboard; public verify portal; QR generation; revocation demo
Phase 1 — MVP	Weeks 4-10	Polygon mainnet deploy; bulk CSV issuance; expiry module; IPFS document backup; pilot with 1 college client
Phase 2 — Beta	Weeks 11-18	Soulbound NFT module; Open Badges 3.0 export; white-label portal; HRMS webhook integration; DigiLocker push
Phase 3 — GA	Month 5-6	Multi-tenant SaaS; verifier B2C API; analytics dashboard; SOC 2 Type I audit readiness; sales to 5+ clients
Phase 4 — Scale	Month 7+	Government tender pursuit; consortium model (multi-university registry); Series A prep; international expansion

10. Team & Resources Required

Role	Responsibilities	Engagement
Blockchain Engineer (Lead)	Solidity contracts, audits, Polygon infra, relayer setup	Full-time

Full-Stack Developer	React dashboard, Node.js API, IPFS integration, bulk issuance	Full-time
Mobile Developer	React Native recipient app, QR scanner, NFT wallet view	Contract / Part-time
DevOps / Infra	Cloud infra (AWS/GCP), CI/CD, RPC node management, DB	Part-time
Product / BD	Client onboarding, pilot coordination, government relations	Full-time

11. Risk Assessment & Mitigation

Risk	Severity	Mitigation
Smart contract vulnerability / exploit	High	Third-party audit (CertiK / Hacken) before mainnet; immutable-by-design registry; no custody of funds
Issuer impersonation (fake university on platform)	High	Manual KYB onboarding; issuer wallet tied to legal entity; IssuerRegistry whitelist enforced at contract level
Legal status of blockchain records in Indian courts	Medium	IT Act Section 65B compliance; cryptographic hash = electronic evidence; design as supplement to official records
Low issuer adoption — institutions prefer paper	Medium	Free Starter tier lowers barrier; NEP 2020 & UGC ABC mandate creates regulatory tailwind; focus on digital-native private colleges first
IPFS link rot (document unavailable)	Low	Verification works without document (hash-only mode); Pinata paid pinning ensures 99.9% uptime; clients can self-host
Polygon network issues / chain migration	Low	Contract designed as upgradable proxy; migration path to zkEVM; all hashes also exportable to CSV as backup

12. Competitive Landscape

CertChain competes in a market currently served by a mix of enterprise incumbents, government platforms, and niche blockchain startups — none of which adequately serves the Indian mid-market.

Competitor	Type	Gap	CertChain Advantage
------------	------	-----	---------------------

DigiLocker (Govt.)	Govt. portal	Issuer-only, no open API	Fully open verification; B2B API; global interoperability
Credly / Accredible	SaaS (US)	No blockchain; USD pricing; no India infra	On-chain tamper-proof; INR pricing; India compliance
MIT Digital Credentials	Open-source	DIY infra; no managed SaaS	Fully managed; zero infra burden for clients
Blockcerts	Open standard	No issuer management; developer-only	Full-stack product; non-technical issuers supported
Entri / SpringVerify	BGV SaaS	Verifier-side only; not blockchain	Issuer-side prevention; cryptographic proof vs. phone calls

13. Success Metrics

13.1 POC Success Criteria

- Full demo: issue certificate on testnet -> generate QR -> verify via public portal -> revoke -> re-verify showing REVOKED — all in under 60 seconds.
- Smart contract unit tests: issueCertificate, verifyCertificate, revokeCertificate all pass with 100% coverage on Hardhat.
- Verification portal returns correct result for valid, tampered, revoked, and expired certificates.
- Bulk CSV of 50 certificates issued in a single batch without error.
- Demo accepted by college evaluators and flagged as production-viable.

13.2 Commercial KPIs (6 months post-launch)

- 3 signed institutional clients (at least 1 university, 1 corporate) live on mainnet.
- 5,000+ certificates issued on Polygon mainnet without a single integrity failure.
- Average verification time under 3 seconds (P99).
- Monthly Recurring Revenue above Rs.1,50,000.
- Zero issuer impersonation incidents.

14. Conclusion & Next Steps

CertChain addresses a high-frequency, high-stakes problem — credential fraud and slow verification — with a technically elegant, commercially scalable solution. The cryptographic

approach is bulletproof, the market need is regulatory-backed (NEP 2020, UGC ABC, IT Act), and the competitive gap in the Indian mid-market is wide open.

The POC phase validates the core on-chain workflow and produces a client-ready, college-presentable demo. The MVP can reach its first paying client within 10 weeks of POC completion.

Recommended immediate next steps:

1. Approve POC scope — assign intern + senior blockchain engineer to Phase 0 (Weeks 1-3).
2. Identify 1-2 pilot clients — target a private engineering college and a 500-person IT company for design-partner onboarding.
3. Register issuer onboarding process — define KYB requirements before any public launch to prevent impersonation risk.
4. File provisional patent — the IssuerRegistry + hash-anchoring architecture may be patentable as a novel application in the Indian context.
5. Begin sales deck — adapt this proposal into a 10-slide investor / client pitch.

CertChain | Confidential & Proprietary | May 2026